Idriss Khattabi

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Location: Available across Morocco



Education

Master in Artificial Intelligence and Data Science — Oct. 2023 – July 2025
Faculty of Sciences and Technologies — Tangier, Morocco

Bachelor in Computer Science and Mathematics — Oct. 2019 – July 2023
Faculty of Poly-disciplinary — Larache, Morocco

Experience

Data / Digitalization Engineer Intern, KOSTAL Group – Tangier — Jan. 2026 – Present

- Analyzed production data to identify bottlenecks, defect patterns, and performance drifts, while exploring AI-driven approaches for anomaly detection, predictive insights, and intelligent process monitoring.
- Built robust Data Pipelines to extract, clean, transform, and consolidate noisy data from multiple operational sources into structured datasets.
- Developed a digital platform for Project Launch Management, improving project tracking, workflow visibility, reporting, and cross-functional collaboration.
- Explored AI-driven approaches for anomaly detection, predictive insights, and intelligent process monitoring using industrial production data.

AI & Data Engineer Intern, Bid2Win – Remote, Part-Time — Dec. 2025 – May 2026

- Developed an AI-powered platform to automate tender analysis, improve document filtering, and support scoring and recommendation workflows.
- Built automated Data Pipelines for scraping, ingesting, cleaning, transforming, chunking, and indexing unstructured tender documents from multiple sources.
- Implemented a RAG-based feature-extraction pipeline to identify tender requirements, eligibility criteria, and relevant insights for scoring and recommendation.

AI & Data Engineer Intern, High Commission for Planning – Remote — Sept. 2025 – Dec. 2025

- Designed and developed an AI-powered platform to automate the documentation and management of statistical processes based on the **GSBPM** framework.
- Engineered document-ingestion and data-processing pipelines to collect, process, structure, and index documents from multiple sources.
- Implemented a RAG engine using **HyDE** prompting and semantic retrieval to generate structured, GSBPM-compliant documentation.
- Automated info-extraction and report-generation workflows, reducing statistical documentation time.

AI & Data Engineer Intern, TNA Consulting – Casablanca — Feb. 2025 – July 2025

- Developed an AI-powered platform enabling financial institutions to search, interpret, and leverage African central-bank publications and regulatory documents.
- Engineered scalable Data Pipelines for scraping, extracting, cleaning, transforming, chunking, embedding, and indexing large volumes of financial documents.
- Built a RAG-based chatbot capable of answering user questions using contextual information retrieved from central-bank publications.

Research Assistant, Faculty of Poly-disciplinary – Larache — Mar. 2023 – June 2023

- Developed an optimization system to improve the consistency of Analytic Hierarchy Process (AHP) pairwise-comparison matrices using Genetic Algorithms (GA) and Particle Swarm Optimization (PSO).
- Built a Django web application for automated AHP matrix generation, validation, and decision-support workflows.
- Co-authored **three** research papers with **Dr. Chakir Tajani**: Paper 1 – Paper 2 – Paper 3

Languages

Arabic: Native proficiency | **English:** Professional (B2) | **French:** Professional (B2)

Certifications

Associate Data Scientist in Python – DataCamp
Associate AI Engineer for Data Scientist – DataCamp
Microsoft Azure Fundamentals – DataCamp
Associate Data Engineer – DataCamp

Skills

- **Programming Languages:** Python, SQL, Java, JavaScript, C, Bash
- **Big Data & Streaming:** Apache PySpark, Apache Kafka, Apache Airflow, RabbitMQ
- **AI & Generative AI:** Machine Learning, Deep Learning, RAG, Prompt Engineering, AI Agents, LangChain, LangGraph, CrewAI, Agno, Ollama, OpenAI API, Fine-Tuning
- **Machine Learning Libraries:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, MLflow
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Snowflake, ChromaDB, Qdrant, Weaviate
- **Backend & APIs:** FastAPI, Django, Flask, Streamlit, REST APIs, Selenium
- **DevOps & Cloud:** Docker, Docker Compose, Git/GitHub, CI/CD, Linux, Azure, AWS, Postman
- **Soft Skills:** Analytical Thinking, Problem Solving, Fast Learning, Teamwork, Communication

Projects

Real-Time Twitter Sentiment Analysis [GitHub Repo]

- Developed a real-time streaming Data Pipeline using Kafka and Spark Streaming to classify tweets.
- Implemented stream-processing workflows for data ingestion, transformation, sentiment classification.
- Built live analytics dashboards to visualize sentiment trends and system performance metrics.
- Tools Used: PySpark, Kafka, Docker, Django, MongoDB, ChartJS.

AI Scraper – AI-Powered Data Extraction Pipeline [GitHub Repo]

- Developed an AI-powered web-scraping application to extract, clean, and structure data from static and dynamic websites using local or cloud-based LLMs.
- Built an automated pipeline to process unstructured web content and transform it into structured datasets for analysis and downstream AI workflows.
- Tools Used: Python, Selenium, Crawl4AI, Streamlit, Ollama, SambaNova API, Pandas.

Data Space for Electronic Medical Records [GitHub Repo]

- Developed a centralized platform for managing, processing, and exploring electronic healthcare data.
- Designed REST APIs and ETL workflows for secure ingestion, transformation, storage, and retrieval of medical records.
- Improved accessibility and interoperability of healthcare data through centralized data features.
- Tools Used: Python, MySQL, REST APIs, ETL Pipelines.

Data Analysis & Machine Learning Tool-Kit Desktop Application [GitHub Repo]

- Developed a Python desktop application that enables users to analyze datasets and apply Machine Learning algorithms through an intuitive graphical interface.
- Implemented data exploration, preprocessing, visualization, and Machine Learning workflows for classification, regression, and clustering tasks.
- Tools Used: Python, Tkinter, Scikit-learn, Pandas, Matplotlib.

ChatPDF – Local RAG Document Assistant [GitHub Repo]

- Developed a local AI assistant that enables users to ask natural-language questions about PDF documents.
- Designed an end-to-end RAG pipeline covering PDF parsing, document chunking, embedding generation, Vector Database retrieval, and LLM-based answer generation.
- Tools Used: Python, Streamlit, LangChain, Ollama, SambaNova API, ChromaDB, Marker.